

Figure 5: Spectrum of $\tilde{\mathbf{H}}$ when training on Cora with SFA.

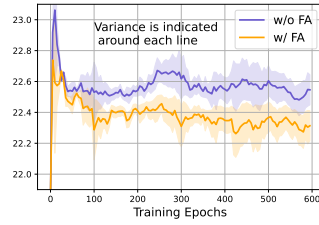


Figure 6: Alignment of $\mathbf{f}$. maps of two views (lower is better).

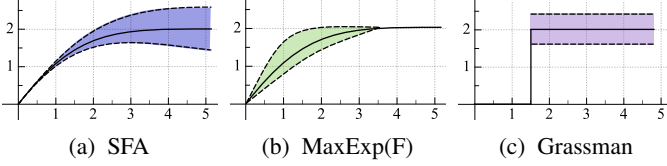


Figure 7: Spectrum augmentation variants (dashed lines: injected noise var.) Power Norm. & Power Norm.* resemble (b) and (a).

along the non-leading singular values (*c.f.* leading singular values of SFA).

$\text{MaxExp}(\mathbf{F})$ is only defined for symmetric-positive definite matrices. We extend it to feature maps by:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{B}(\mathbf{I} - (\mathbf{I} - \mathbf{A}/\text{Tr}(\mathbf{A}))^{\eta+\Delta\eta}), \quad (7)$$

where $\mathbf{A} = (\mathbf{H}^\top \mathbf{H})^{0.5}$ and $\mathbf{B} = (\mathbf{H}^\top \mathbf{H})^{-0.5}$ are obtained via highly-efficient Newton-Schulz iteration. We draw the random noise $\Delta\eta \sim \alpha\mathcal{N}(0, 1)$ for each $\mathbf{H}$ and control the variance by $\alpha > 0$. We round $\eta + \Delta\eta$ to the nearest integer ($\text{MaxExp}(\mathbf{F})$ requires an integer for fast matrix exponentiation) and ensure the integer is greater than zero. See the ?? section for details.

Matrix Square Root is similar to $\text{MaxExp}(\mathbf{F})$ but the soft function flattening the spectrum is replaced with a matrix square root push-forward function. *Power Norm.* yields:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{B} \text{PowerNorm}(\mathbf{A}; \beta + \Delta\beta), \quad (8)$$

where $0 \leq \beta + \Delta\beta \leq 1$ controls the steepness of the balancing push-forward curve. $\Delta\beta$ is drawn from the Normal distribution (we choose the best aug. variance). *Power Norm.** is:

$$\begin{aligned} \tilde{\mathbf{H}} &= \mathbf{H}\mathbf{B}((1 - \beta - \Delta\beta)\mathbf{A}^{0.5} + (\beta + \Delta\beta)\mathbf{A}^{0.5}\mathbf{A}^{0.25}) \\ &= \mathbf{H}\mathbf{B} \text{PowerNorm}(\mathbf{A}; \beta + \Delta\beta), \end{aligned} \quad (9)$$

where $0 \leq 1 + \Delta\beta^* \leq 2$, $\Delta\beta^*$ is drawn from the Normal dist. $\mathbf{A}^{0.5}$ and $\mathbf{A}^{0.25} = (\mathbf{A}^{0.5})^{0.5}$ are obtained by Newton-Schulz iterations. See ?? for details.

Grassman feature maps can be obtained as:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{B}(\mathbf{V}\text{Flat}(\mathbf{A}; \kappa)\mathbf{V}^\top), \quad (10)$$

where $\text{Flat}(\cdot; \kappa)$ sets $\kappa > 0$ leading singular values to $1 + \Delta\kappa$ (the rest is 0), and $\Delta\kappa \sim \alpha\mathcal{N}(0, \mathbf{I})$. In experiments we use SVD and random SVD (injects itself some noise) to produce $\mathbf{V}$ and $\mathbf{A}$ from $\mathbf{A}$. See ?? for details.

Table 10 shows that SFA outperforms $\text{MaxExp}(\mathbf{F})$, Power Norm., Power Norm.* and Grassman. As SFA augments parts of spectrum where signal resides (leading singular values) which is better than augmenting non-leading singular values

Power Iteration	Am-Computer	Cora	CiteSeer
$k = 0$	87.25	83.04	71.35
$k = 1$	88.83	85.90	73.56
$k = 2$	88.72	85.59	75.3
$k = 4$	88.64	85.29	74.8
$k = 8$	88.27	85.23	74.9

Table 8: Results on different k of SFA.

ϵ		10^{-4}	10^{-3}	10^{-2}	10^{-1}
Cora	w/o SFA	80.55	70.67	62.85	59.23
	with SFA	83.14	76.56	69.59	62.55
CiteSeer	w/o SFA	67.48	62.32	52.69	42.81
	with SFA	72.12	70.36	60.40	46.44

Table 9: Results on noisy features.

Spectrum Aug.	Am-Comput.	Cora	CiteSeer
SFA (ours)	88.83	85.90	75.3
MaxExp(F)	87.13	84.19	72.85
MaxExp(F) (w/o noise)	86.83	83.52	72.35
Power Norm.*	86.7	82.9	70.4
Power Norm.	86.6	82.7	70.2
Power Norm. (w/o noise)	86.3	82.5	69.9
Grassman	86.23	82.57	70.15
Grassman (w/o noise)	85.83	81.17	69.85
Grassman (rand. SVD)	85.33	81.01	69.95
Matrix Precond.	82.52	78.14	67.73

Table 10: Results of various spectral augmentations.

(some noise may reside there) as in $\text{MaxExp}(\mathbf{F})$. As Grassman binarizes spectrum, it may reject some useful signal at the boundary between leading and non-leading singular values. Finally, ?? model reduces the spectral gap, thus rebalances the spectrum (also non-leading part).

5 Conclusions

We have shown that GCL is not restricted to only link perturbations or feature augmentation. By introducing a simple and efficient spectral feature augmentation layer we achieve significant performance gains. Our incomplete power iteration is very fast. Our theoretical analysis has demonstrated that SFA rebalances the useful part of spectrum, and also augments the useful part of spectrum by implicitly injecting the noise into singular values of both data views. SFA leads to a better alignment with a lower generalization bound.

6 Acknowledgments

We thank anonymous reviewers for their valuable comments. The work described here was partially supported by grants from the National Key Research and Development Program of China (No. 2018AAA0100204) and from the Research Grants Council of the Hong Kong Special Administrative Region, China (CUHK 2410021, Research Impact Fund, No. R5034-18). Minus temporibus veritatis adipisci quia blanditiis a nisi earum tenetur, necessitatibus molestias iure quas incident aliquam sunt in, possimus placeat nostrum voluptatum, repellendus commodi omnis debitis qui pariat officis rem, beatae nam cupiditate sunt ut. Itaque unde vitae a, quam quibusdam repellat quae laboriosam neque ullam voluptatem pariat tenetur, aperiam molestiae veritatis laudantium quos deleniti explicabo, eveniet consequuntur quidem provident voluptas neque quibusdam totam distinctio iste deserunt, ipsa blanditiis aliquid corporis? Beatae commodi eum dolores alias

temporibus, magni expedita id nam sint autem corporis quia voluptates ipsam suscipit, natus in sapiente veritatis officia vero vel atque fuga, dolor esse quae mollitia reprehenderit.